

PIL'VO, Russian Federation

WMO: 320690

Lat: 50.0462N

Lon: 142.1735E

Elev: 4

StdP: 101.28

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-24.5	-28.7	0.3	-25.9	-26.7	0.3	-24.1	15.9	-9.4	14.1	-9.7	4.7	120	0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	22.1	18.4	20.9	17.9	19.8	17.3	19.4	21.2	18.6	20.2	17.7	19.4	4.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.5	20.6	17.8	12.8	19.7	16.9	12.1	19.0	55.6	21.3	52.9	20.3	50.2	19.4	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.5	11.6	9.5	-29.4	24.4	2.7	1.3	-31.4	25.4	-33.0	26.2	-34.5	26.9	-36.4	27.9		
			DB	WB	-29.6	20.7	2.7	1.6	-31.5	21.8	-33.0	22.8	-34.5	23.7	-36.4	24.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-14.8	-14.1	-7.8	-0.1	5.7	10.7	15.1	16.4	12.3	5.1	-3.4	-11.4
	DBStd	11.56	5.23	4.89	4.55	3.36	3.04	2.65	2.30	2.35	3.11	3.34	4.55	5.12
	HDD10.0	3697	769	674	552	303	139	23	1	0	13	155	403	664
	HDD18.3	6257	1027	908	810	553	393	229	101	69	181	410	653	923
	CDD10.0	493	0	0	0	0	5	44	160	198	83	3	0	0
	CDD18.3	12	0	0	0	0	0	0	2	9	1	0	0	0
	CDH23.3	8	0	0	0	0	0	0	1	8	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.2	4.6	4.4	4.2	4.1	3.8	3.4	3.1	3.4	4.1	4.6	5.1	5.0
Precipitation	PrecAvg	702	36	23	32	43	54	49	62	87	109	95	63	48
	PrecMax	954	98	59	68	106	123	131	165	222	180	196	129	90
	PrecMin	492	7	3	2	17	16	8	0	8	27	21	8	23
	PrecStd	95	19	14	18	21	28	27	34	41	41	47	25	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	1.1	4.2	12.4	17.1	20.1	22.6	24.3	21.7	16.3	9.8	2.2
		MCWB	-1.1	-0.5	1.3	6.2	10.7	14.3	18.1	20.3	17.1	12.4	7.1	1.0
	2%	DB	-1.9	-1.9	2.1	8.7	14.5	18.0	21.1	22.9	20.2	14.3	6.8	0.5
		MCWB	-2.9	-3.0	0.1	4.6	9.9	13.8	17.9	19.4	16.7	11.1	4.7	-0.6
	5%	DB	-4.4	-3.9	0.7	6.4	12.4	16.4	19.9	21.7	19.0	12.6	4.5	-1.4
		MCWB	-5.3	-4.9	-1.0	3.5	8.7	13.3	17.2	18.6	16.1	9.8	2.7	-2.5
	10%	DB	-6.7	-5.8	-0.6	4.7	10.7	14.9	18.8	20.6	17.9	11.1	2.7	-3.6
		MCWB	-7.3	-6.6	-2.0	2.3	7.9	12.5	16.7	18.1	15.2	8.5	1.2	-4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-0.4	1.8	7.3	12.5	16.0	19.3	21.5	18.7	13.5	7.6	1.2
		MCDB	0.2	1.1	3.3	11.0	15.5	18.4	21.2	23.5	20.6	15.5	9.2	2.1
	2%	WB	-2.9	-2.8	0.3	5.1	10.4	14.7	18.5	20.2	17.5	11.6	4.9	-0.5
		MCDB	-1.7	-1.9	1.7	8.1	13.4	16.9	20.3	22.0	19.3	13.7	6.5	0.4
	5%	WB	-5.2	-4.8	-0.8	3.7	9.3	13.6	17.7	19.2	16.6	10.1	2.9	-2.5
		MCDB	-4.5	-4.0	0.6	6.1	11.8	15.7	19.3	20.9	18.5	12.2	4.4	-1.5
	10%	WB	-7.3	-6.6	-2.0	2.5	8.2	12.7	16.9	18.5	15.5	8.8	1.2	-4.6
		MCDB	-6.6	-5.8	-0.7	4.4	10.2	14.5	18.5	20.1	17.6	10.8	2.3	-3.6
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.5	8.6	6.1	6.1	5.6	5.0	6.1	7.9	7.0	5.9	7.0
		MCDBR	7.5	9.3	8.8	9.3	9.7	8.5	6.5	7.3	8.8	9.6	7.6	7.3
		MCWBR	7.0	8.5	7.4	6.0	5.9	5.2	4.0	4.8	6.3	7.4	6.3	6.7
	5% WB	MCDBR	7.5	8.7	8.3	8.4	8.2	7.0	5.2	5.7	7.5	9.0	7.5	7.1
		MCWBR	6.9	8.0	7.2	5.9	5.5	4.8	3.6	4.3	5.9	7.5	6.5	6.7
Clear-Sky Solar Irradiance	taub	0.298	0.333	0.406	0.507	0.548	0.506	0.476	0.442	0.396	0.412	0.371	0.311	
	taud	2.047	1.942	1.817	1.762	1.801	2.011	2.170	2.284	2.340	2.096	2.044	2.049	
	Ebn at Noon	699	765	766	732	724	764	784	794	793	673	593	611	
	Edh at Noon	87	125	171	203	204	167	141	120	102	108	89	76	
All-Sky Solar Radiation	RadAvg	1.35	2.47	3.67	4.34	4.84	5.29	4.74	4.15	3.45	2.14	1.20	0.95	
	RadStd	0.12	0.16	0.27	0.44	0.46	0.50	0.54	0.38	0.30	0.17	0.12	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	+0.35	+1.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page